

Enhancing Learning Outcomes and Reducing Anxiety in Programming Courses through AI Integration

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Abstract:

This study aims to explore whether integrating AI into programming courses can enhance learning outcomes and reduce learning anxiety in the context of widespread AI applications. Beginner programming students often encounter situations where their code structure is correct, but the output does not meet the requirements, primarily due to misunderstandings of the problem or logical errors in the code. To address this challenge, a virtual learning partner interface was developed on the Line platform, using simple and clear explanations to lower the learning barrier for programming. Additionally, key shortcut keywords were designed to allow the AI to automatically generate three related questions after a brief response, promoting active learning and critical thinking. The system also incorporates ChatGPT for debugging and logical correction of code, providing better programming suggestions.

The study involved 90 first-year programming beginners at a university in central Taiwan. It examined the impact of the virtual learning partner system on learning outcomes and programming anxiety. The experimental content covered basic Python programming, logical judgment, loop processing, big data analysis, and chart creation. The study used Venkatesh's (2000) Learning Anxiety Scale, administered before and after learning, to assess the effectiveness of the virtual learning partner.

Results showed a significant improvement in students' learning outcomes after AI was introduced into the programming course. The average score difference was 33.23622, with a standard deviation of 26.67530, corresponding to a t-value of 11.820 and a p-value of 0.000, indicating a significant difference between pre- and post-test scores. Additionally, the anxiety measurement results indicated a mean difference of 2.62222, with a t-value of 7.626 and a p-value of 0.000, confirming that the virtual learning partner effectively reduced programming learning anxiety.

Keywords: *Artificial Intelligence in Education, Programming Learning Anxiety, Virtual Learning Partner*



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In 2023, I was studying in the doctoral program at the Graduate Institute of Educational Information and Measurement at National Taichung University of Education in Taiwan. My research focuses on applying AI tools in the classroom.